THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{9}$	List all pairs with the given product.
2	**(d)** $\frac{2}{9}$	Count the favourable pairs out of 36.
3	**(b)** $\frac{7}{8}$	Use the complement rule where it is quicker.
4	**(a)** 4	Let blue $= x$ and solve $\frac{x}{r+x} = \frac{p}{q}$.
5	**(c)** 21	Blue = total − red.
6	**(d)** $\frac{7}{11}$	Count with repetition.
7	**(a)** 600	Expected failures $= N(1-p)$.
8	**(b)** 0.80	Reason about probabilities.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$P = \frac{3}{6} = \frac{1}{2}$, obtained exactly as in R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R is a general range, so it does not explain why an impossible event gets exactly 0.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Favourable pairs: (2,6), (3,4), (4,3), (6,2) (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1
12	Let blue $= x$: $\frac{x}{12+x} = \frac{1}{4}$; $4x = 12 + x$ (1); $x = 4$ (1).	1+1
13	$P(\text{no head}) = P(\text{TTT}) = \frac{1}{8}$ (1); $P(\text{at least one head}) = 1 - \frac{1}{8} = \frac{7}{8}$ (1).	1+1
14	Favourable pairs: (1,6), (2,3), (3,2), (6,1) (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{green}) = 1 - \frac{1+1}{3} = \frac{1}{3}$; Red $= 8$ (1); Blue $= 8$ (1); Green $= 8$ (1).	1+1+1
16	$P(\text{prime}) = \frac{52}{100} = \frac{13}{25}$; $P(>4) = \frac{31}{100} = \frac{31}{100}$ (1); Either $= 52 + 31 - 14 = 69$ (1); $P = \frac{69}{100} = \frac{69}{100}$ (1).	1+1+1
17	Number of letters $= 11$; $P(\text{not } S) = 1 - \frac{4}{11} = \frac{7}{11}$ (1); Vowel or S: 8 letters (1); $P = \frac{8}{11}$ (1).	1+1+1
	OR: $P(\text{green}) = 1 - \frac{1+2}{5} = \frac{2}{5}$; Red $= 8$ (1); Blue $= 16$ (1); Green $= 16$ (1).	1+1+1
18	$P(\text{prime}) = \frac{108}{200} = \frac{27}{50}$; $P(>4) = \frac{63}{200} = \frac{63}{200}$ (1); Either $= 108 + 63 - 30 = 141$ (1); $P = \frac{141}{200} = \frac{141}{200}$ (1).	1+1+1
	OR: Number of letters $= 11$; $P(\text{not } T) = 1 - \frac{2}{11} = \frac{9}{11}$ (1); Vowel or T: 6 letters (1); $P = \frac{6}{11}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 6 outcomes: 1, 2, 3, 4, 5, 6 (1). (ii) $\frac{3}{6} = \frac{1}{2}$ (1). (iii) Favourable outcomes 5, 6, so $\frac{2}{6} = \frac{1}{3}$ (1). (iv) Multiples of 3 are 3 and 6, so $P = \frac{1}{3}$ (1); $P(\text{not}) = 1 - \frac{1}{3} = \frac{2}{3}$ (1).	1+1+1+2
20	Sum 7: 6 outcomes; more than 7: 15 (1); $P(\text{sum} = 7) = \frac{1}{6}$ (1); $P(>7) = \frac{5}{12}$ (1); $P(\leq 7) = 1 - \frac{5}{12} = \frac{7}{12}$ (1); The two probabilities add to 1 (1).	1+1+1+1+1
	OR: Let x be removed: $\frac{15-x}{33-x} = \frac{1}{4}$ (1); $4(15-x) = 1(33-x)$ (1); $60 - 4x = 33 - x$ (1); $x = 9$ (2).	1+1+1+2

Mistakes to avoid:

- Giving a probability greater than 1 or a negative value, or leaving it as a count instead of a fraction.
- Using the number of favourable outcomes divided by the number of *unfavourable* outcomes instead of the *total* outcomes.
- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{3}{4}$	Use the complement rule where it is quicker.
2	**(c)** 15	Let blue $= x$ and solve $\frac{x}{r+x} = \frac{p}{q}$.
3	**(c)** 27	Blue = total − red.
4	**(b)** $\frac{6}{11}$	Count with repetition.
5	**(b)** 700	Expected failures $= N(1-p)$.
6	**(b)** $\frac{1}{3}$	Reason about probabilities.
7	**(d)** $\frac{1}{12}$	List all pairs with the given product.
8	**(d)** $\frac{7}{36}$	Count the favourable pairs out of 36.
9	**(c)** A is true but R is false.	A is correct; R is false because the sum is 1.
10	**(d)** A is false but R is true.	A is false because probability never exceeds 1; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let blue $= x$: $\frac{x}{15+x} = \frac{2}{5}$; $5x = 30 + 2x$ (1); $x = 10$ (1).	1+1
12	Outcomes with maximum 3: 5 of 36 (1); $P = \frac{5}{36}$ (1).	1+1
13	Favourable pairs: (2,4), (4,2) (1); $P = \frac{2}{36} = \frac{1}{18}$ (1).	1+1
14	Let blue $= x$: $\frac{x}{15+x} = \frac{3}{8}$; $8x = 45 + 3x$ (1); $x = 9$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{green}) = 1 - \frac{2+3}{8} = \frac{3}{8}$; Red = 12 (1); Blue = 18 (1); Green = 18 (1).	1+1+1
16	$P(\text{prime}) = \frac{80}{150} = \frac{8}{15}$; $P(>4) = \frac{46}{150} = \frac{23}{75}$ (1); Either $= 80 + 46 - 22 = 104$ (1); $P = \frac{104}{150} = \frac{52}{75}$ (1).	1+1+1
17	Number of letters = 11; $P(\text{not } N) = 1 - \frac{2}{11} = \frac{9}{11}$ (1); Vowel or N: 8 letters (1); $P = \frac{8}{11}$ (1).	1+1+1
	OR: $P(\text{green}) = 1 - \frac{1+3}{6} = \frac{1}{3}$; Red = 6 (1); Blue = 18 (1); Green = 12 (1).	1+1+1
18	$P(\text{prime}) = \frac{66}{120} = \frac{11}{20}$; $P(>4) = \frac{38}{120} = \frac{19}{60}$ (1); Either $= 66 + 38 - 18 = 86$ (1); $P = \frac{86}{120} = \frac{43}{60}$ (1).	1+1+1
	OR: Number of letters = 11; $P(\text{not } B) = 1 - \frac{2}{11} = \frac{9}{11}$ (1); Vowel or B: 6 letters (1); $P = \frac{6}{11}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{90}{200} = \frac{9}{20}$ (1). (ii) $\frac{70}{200} = \frac{7}{20}$ (1). (iii) $\frac{40}{200} = \frac{1}{5}$ (1). (iv) $1 - \frac{9}{20} = \frac{11}{20}$ (1); expected UPI customers $= 500 \times \frac{7}{20} = 175$ (1).	1+1+1+2
20	Sum 9: 4 outcomes; more than 9: 6 (1); $P(\text{sum} = 9) = \frac{1}{9}$ (1); $P(>9) = \frac{1}{6}$ (1); $P(\leq 9) = 1 - \frac{1}{6} = \frac{5}{6}$ (1); The two probabilities add to 1 (1).	1+1+1+1+1
	OR: Let x be removed: $\frac{12-x}{26-x} = \frac{1}{3}$ (1); $3(12-x) = 1(26-x)$ (1); $36 - 3x = 26 - x$ (1); $x = 5$ (2).	1+1+1+2

Mistakes to avoid:

- Giving a probability greater than 1 or a negative value, or leaving it as a count instead of a fraction.
- Using the number of favourable outcomes divided by the number of *unfavourable* outcomes instead of the *total* outcomes.
- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 18	Blue = total − red.
2	**(d)** $\frac{7}{11}$	Count with repetition.
3	**(d)** 1800	Expected failures $= N(1-p)$.
4	**(a)** 1	Reason about probabilities.
5	**(b)** $\frac{1}{36}$	List all pairs with the given product.
6	**(d)** $\frac{1}{4}$	Count the favourable pairs out of 36.
7	**(a)** $\frac{1}{2}$	Use the complement rule where it is quicker.
8	**(b)** 12	Let blue $= x$ and solve $\frac{x}{r+x} = \frac{p}{q}$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$P = \frac{2}{4} = \frac{1}{2}$, from the sample space in R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$\frac{54}{100} = 0.54$; both are true, but R does not give the experimental value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Outcomes with maximum 5: 9 of 36 (1); $P = \frac{1}{4}$ (1).	1+1
12	Favourable pairs: (1,5), (5,1) (1); $P = \frac{2}{36} = \frac{1}{18}$ (1).	1+1
13	Let blue $= x$: $\frac{x}{18+x} = \frac{4}{7}$; $7x = 72 + 4x$ (1); $x = 24$ (1).	1+1
14	Outcomes with maximum 6: 11 of 36 (1); $P = \frac{11}{36}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{green}) = 1 - \frac{3+4}{12} = \frac{5}{12}$; Red $= 15$ (1); Blue $= 20$ (1); Green $= 25$ (1).	1+1+1
16	$P(\text{prime}) = \frac{41}{80} = \frac{41}{80}$; $P(>4) = \frac{25}{80} = \frac{5}{16}$ (1); Either $= 41 + 25 - 11 = 55$ (1); $P = \frac{55}{80} = \frac{11}{16}$ (1).	1+1+1
17	Number of letters $= 10$; $P(\text{not } H) = 1 - \frac{1}{5} = \frac{4}{5}$ (1); Vowel or H: 5 letters (1); $P = \frac{1}{2}$ (1).	1+1+1
	OR: $P(\text{green}) = 1 - \frac{2+5}{9} = \frac{2}{9}$; Red $= 12$ (1); Blue $= 30$ (1); Green $= 12$ (1).	1+1+1
18	$P(\text{prime}) = \frac{130}{250} = \frac{13}{25}$; $P(>4) = \frac{84}{250} = \frac{42}{125}$ (1); Either $= 130 + 84 - 39 = 175$ (1); $P = \frac{175}{250} = \frac{7}{10}$ (1).	1+1+1
	OR: Number of letters $= 9$; $P(\text{not } A) = 1 - \frac{1}{3} = \frac{2}{3}$ (1); Vowel or A: 3 letters (1); $P = \frac{1}{3}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\{HH, HT, TH, TT\}$ (1). (ii) $\frac{1}{4}$ (1). (iii) $\frac{3}{4}$ (1). (iv) $P(\text{no head}) = \frac{1}{4}$ and $1 - \frac{1}{4} = \frac{3}{4}$ (1); $P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2}$, so expected $= 80 \times \frac{1}{2} = 40$ times (1).	1+1+1+2
20	Sum 10: 3 outcomes; more than 10: 3 (1); $P(\text{sum} = 10) = \frac{1}{12}$ (1); $P(>10) = \frac{1}{12}$ (1); $P(\leq 10) = 1 - \frac{1}{12} = \frac{11}{12}$ (1); The two probabilities add to 1 (1).	1+1+1+1+1
	OR: Let x be removed: $\frac{14-x}{30-x} = \frac{1}{5}$ (1); $5(14-x) = 1(30-x)$ (1); $70 - 5x = 30 - x$ (1); $x = 10$ (2).	1+1+1+2

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- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

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